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Exercise list: L08 — Univariate time series (Class 8): textbook exercises

Instructions. These are the *Practice* items of Class 8 in `Map/class-plan.md` (T 2.1–2.6; W 18.C13; T 3.3, 3.4, 3.6, 3.11, 3.12, 3.14; W 12.4), grouped by book. Every question opens with a source line, and the statement is reproduced verbatim from the book PDF. Data exercises use Tsay's data files (FTS 3e web page, retrieved from the Internet Archive) and the `wooldridge` package; see `Listas/data/README.md`. They are solved in Python (`statsmodels`; GARCH-type models by conditional maximum likelihood written in NumPy/SciPy); `Listas/L08-book-exercises-check.py` reproduces and asserts every number in the solutions. Tests use the 5% level unless the statement says otherwise. Show your steps.

Block 1 — Tsay, Chapter 2: linear time series *(T Ch. 2 • Questions 1–6)*

Question 1. Tsay, Exercise 2.1 (MA(1) forecasts).

Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 2, Exercise 2.1, printed p. 104, reproduced verbatim. The chapter's exercises open with: "If not specifically specified, use 5% significance level to draw conclusions in the exercises."

2.1. Suppose that the simple return of a monthly bond index follows the MA(1) model

$$R_t = a_t + 0.2a_{t-1}, \qquad \sigma_a = 0.025.$$

Assume that $a_{100} = 0.01$. Compute the 1-step- and 2-step-ahead forecasts of the return at the forecast origin $t = 100$. What are the standard deviations of the associated forecast errors? Also compute the lag-1 and lag-2 autocorrelations of the return series.

Solution:

Question 2. Tsay, Exercise 2.2 (AR(2) with $\phi_1 = 0$: moments and forecasts).

Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 2, Exercise 2.2, printed p. 104, reproduced verbatim.

2.2. Suppose that the daily log return of a security follows the model

$$r_t = 0.01 + 0.2r_{t-2} + a_t,$$

where $\{a_t\}$ is a Gaussian white noise series with mean zero and variance 0.02. What are the mean and variance of the return series r_t? Compute the lag-1 and lag-2 autocorrelations of r_t. Assume that $r_{100} = -0.01$, and $r_{99} = 0.02$. Compute the 1- and 2-step-ahead forecasts of the return series at the forecast origin $t = 100$. What are the associated standard deviations of the forecast errors?

Solution:

Question 3. Tsay, Exercise 2.3 (U.S. unemployment rate: model, forecasts, business cycles).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 2, Exercise 2.3, printed p. 104, reproduced verbatim.

2.3. Consider the monthly U.S. unemployment rate from January 1948 to March 2009 in the file `m-unrate.txt`. The data are seasonally adjusted and obtained from the Federal Reserve Bank of St Louis. Build a time series model for the series and use the model to forecast the unemployment rate for the April, May, June, and July of 2009. In addition, does the fitted model imply the existence of business cycles? Why? (Note that there are more than one model fits the data well. You only need an adequate model.)

Solution:

Question 4. Tsay, Exercise 2.4 (CRSP decile returns: Ljung–Box test, ARMA model, forecasts).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 2, Exercise 2.4, printed p. 105, reproduced verbatim. The data file (not named in the statement) is Tsay's `m-deciles08.txt`.

2.4. Consider the monthly simple returns of the Decile 1, Decile 2, Decile 9, and Decile 10 of NYSE/AMEX/NASDAQ based on market capitalization. The data span is from January 1970 to December 2008, and the data are obtained from CRSP.

(a) For the return series of Decile 2 and Decile 10, test the null hypothesis that the first 12 lags of autocorrelations are zero at the 5% level. Draw your conclusion.

(b) Build an ARMA model for the return series of Decile 2. Perform model checking and write down the fitted model.

(c) Use the fitted ARMA model to produce 1- to 12-step-ahead forecasts of the series and the associated standard errors of forecasts.

Solution:

Question 5. Tsay, Exercise 2.5 (long-range dependence in $|R_t|$ of IBM).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 2, Exercise 2.5, printed p. 105, reproduced verbatim.

2.5. Consider the daily simple returns of IBM stock from 1970 to 2008 in the file `d-ibm3dx7008.txt`. Compute the first 100 lags of ACF of the absolute series of daily simple returns of IBM stock. Is there evidence of long-range dependence? Why?

Solution:

Question 6. Tsay, Exercise 2.6 (seasonal model for electricity demand).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 2, Exercise 2.6, printed p. 105, reproduced verbatim.

2.6. Consider the demand of electricity of a manufacturing sector in the United States. The data are logged, denote the demand of a fixed day of each month, and are in `power6.txt`. Build a time series model for the series and use the fitted model to produce 1- to 24-step-ahead forecasts.

Solution:

Block 2 — Tsay, Chapter 3: conditional heteroscedastic models *(T Ch. 3 • Questions 7–12)*

Question 7. Tsay, Exercise 3.3 (Gaussian AR(1)–GARCH(1,1) likelihood).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 3, Exercise 3.3, printed p. 168, reproduced verbatim.

3.3. Suppose that $r_1, \ldots, r_n$ are observations of a return series that follows the AR(1)-GARCH(1,1) model

$$r_t = \mu + \phi_1 r_{t-1} + a_t, \qquad a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where ϵ_t is a standard Gaussian white noise series. Derive the conditional log-likelihood function of the data.

Solution:

Question 8. Tsay, Exercise 3.4 (Student-t AR(1)–GARCH(1,1) likelihood).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 3, Exercise 3.4, printed p. 168, reproduced verbatim.

3.4. In the equation in Exercise 3.3, assume that ϵ_t follows a standardized Student-t distribution with v degrees of freedom. Derive the conditional log-likelihood function of the data.

Solution:

Question 9. Tsay, Exercise 3.6 (Merck monthly returns: serial correlation, ARCH effects, ARCH model).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 3, Exercise 3.6, printed p. 169, reproduced verbatim.

3.6. The file `m-mrk4608.txt` contains monthly simple returns of Merck stock from June 1946 to December 2008. The file has two columns denoting date and simple return. Transform the simple returns to log returns.

(a) Is there any evidence of serial correlations in the log returns? Use autocorrelations and 5% significance level to answer the question. If yes, remove the serial correlations.

(b) Is there any evidence of ARCH effects in the log returns? Use the residual series if there are serial correlations in part (a). Use Ljung–Box statistics for the squared returns (or residuals) with 6 and 12 lags of autocorrelations and 5% significance level to answer the question.

(c) Identify an ARCH model for the data and fit the identified model. Write down the fitted model.

Solution:

Question 10. Tsay, Exercise 3.11 (GM daily log returns: ARCH test, PACF, normal and GED GARCH).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 3, Exercise 3.11, printed p. 170, reproduced verbatim.

3.11. The file `d-gmsp9908.txt` contains the daily simple returns of GM stock and the S&P composite index from 1999 to 2008. It has three columns denoting date, GM return, and S&P return.

(a) Compute the daily log returns of GM stock. Is there any evidence of ARCH effects in the log returns? You may use 10 lags of the squared returns and 5% significance level to perform the test.

(b) Compute the PACF of the squared log returns (10 lags).

(c) Specify a GARCH model for the GM log return using a normal distribution for the innovations. Perform model checking and write down the fitted model.

(d) Find an adequate GARCH model for the series but using the generalized error distribution for the innovations. Write down the fitted model.

Solution:

Question 11. Tsay, Exercise 3.12 (S&P daily simple returns: ARCH test, GARCH model, forecasts).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 3, Exercise 3.12, printed p. 170, reproduced verbatim.

3.12. Consider the daily simple returns of the S&P composite index in the file `d-gmsp9908.txt`.

(a) Is there any ARCH effect in the simple return series? Use 10 lags of the squared returns and 5% significance level to perform the test.

(b) Build an adequate GARCH model for the simple return series.

(c) Compute 1-step- to 4-step-ahead forecasts of the simple return and its volatility based on the fitted model.

Solution:

Question 12. Tsay, Exercise 3.14 (market volatility as a regressor in GM's volatility).
Source: Tsay (2010), *Analysis of Financial Time Series*, 3rd ed., Chapter 3, Exercise 3.14, printed p. 171, reproduced verbatim.

3.14. Revisit the file `d-gmsp9908.txt`. However, we shall investigate the value of using market volatility in modeling volatility of individual stocks. Convert the two simple return series into percentage log return series.

(a) Build an AR(5)–GARCH(1,1) model with generalized error distribution for the log S&P returns. The AR(5) contains only lags 3 and 5. Denote the fitted volatility series by `spvol`.

(b) Estimate a GARCH(1,1) model with `spvol` as an exogenous variable to the log GM return series. Check the adequacy of the model, and write down the fitted model. In S-Plus, the command is

```
fit = garch(gm ~1, ~garch(1,1)+spvol, cond.dist='ged')
```

(c) Discuss the implication of the fitted model.

Solution:

Block 3 — Wooldridge: unit roots and ARCH *(W Ch. 18 and 12 • Questions 13–14)*

Question 13. Wooldridge, Computer Exercise 18.C13 (Dickey–Fuller tests, TRAFFIC2).
Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 18, Computer Exercise C13, printed p. 604, reproduced verbatim.

C13 Use the data in TRAFFIC2 for this exercise. These monthly data, on traffic accidents in California over the years 1981 to 1989, were used in Computer Exercise C11 in Chapter 10.

(i) Using the standard Dickey-Fuller regression, test whether $ltotacc_t$ has a unit root. Can you reject a unit root at the 2.5% level?

(ii) Now, add two lagged changes to the test from part (i) and compute the augmented Dickey-Fuller test. What do you conclude?

(iii) Add a linear time trend to the ADF regression from part (ii). Now what happens?

(iv) Given the findings from parts (i) through (iii), what would you say is the best characterization of $ltotacc_t$: an I(1) process or an I(0) process about a linear time trend?

(v) Test the percentage of fatalities, $prcfat_t$, for a unit root, using two lags in an ADF regression. In this case, does it matter whether you include a linear time trend?

Solution:

Question 14. Wooldridge, Problem 12.4 (ARCH and serial correlation).

Source: Wooldridge (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Chapter 12, Problem 4, printed p. 397, reproduced verbatim.

4 True or false: "If the errors in a regression model contain ARCH, they must be serially correlated."

Solution: